

UNIT-III

FOURIER SERIES

Topic Learning Objectives:

Upon Completion of this unit, students will be able to:

- Understand the periodic phenomena.
- Develop the knowledge of periodic functions as a Fourier series subject to Dirichlet's conditions and derive the Fourier series using Euler's formulae.
- Expand a function in complex form of Fourier series,
- Represent continuous-time periodic signals using Fourier series.
- Find sine and cosine series.

1. Introduction:

French mathematician and physicist Joseph Fourier for the first time used Fourier series in his work in which he developed the theory of heat conduction. These series had a deep influence in the further development of mathematics and mathematical physics. In many engineering problems, especially in the study of periodic phenomena in the conduction of heat, electrodynamics and acoustics it is necessary to express a function in a series of sines and cosines. Fourier series are infinite series designed to represent general periodic functions in terms of cosines and sines. Fourier series are very important to engineers and physicist as they allow the solution of ODEs in connection with forced oscillations and the approximation of periodic functions. Fourier series are more universal than the Taylor series in calculus as many *discontinuous* periodic functions in applications that do not have Taylor series expansions can be developed in Fourier series.



“JEAN-BAPTISTE JOSEPH FOURIER (1768–1830), French physicist and mathematician, lived and taught in Paris, accompanied Napoléon in the Egyptian War and was later made prefect of Grenoble. The beginnings on Fourier series can be found in works by Euler and by Daniel Bernoulli, but it was Fourier who employed them in a systematic and general manner in his main work, *Théorie analytique de la chaleur* (Analytic Theory of Heat, Paris, 1822), in which he developed the theory of heat conduction making these series a most important tool in applied mathematics.”

Periodic function

Definition: A general periodic function $f(x)$ of period T satisfies the condition

$$f(x + T) = f(x).$$

Here $f(x)$ is a real-valued function and $T > 0$ is a positive real number.

As a consequence, it follows that

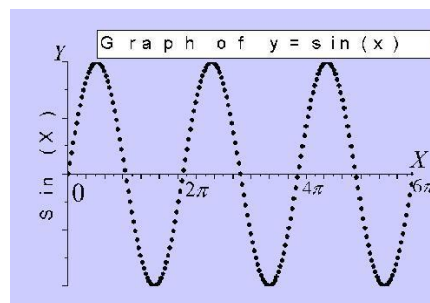
$$f(x) = f(x + T) = f(x + 2T) = f(x + 3T) = \dots = f(x + nT).$$

Thus,

$$f(x) = f(x + nT), n=1,2, 3,\dots$$

The function $f(x) = \sin x$ is periodic of period 2π since $\sin(x+2n\pi) = \sin x$, $n=1, 2, 3,\dots$

The graph of the function is shown below:



Note that the graph of the function between 0 and 2π is the same as that between 2π and 4π and so on. It may be verified that a linear combination of periodic functions is also periodic.

Examples: $\cos x$, $\sin x$, $\sec x$, $\operatorname{cosec} x$, are periodic functions with period 2π .

$\tan x$, $\cot x$ are periodic with period π .

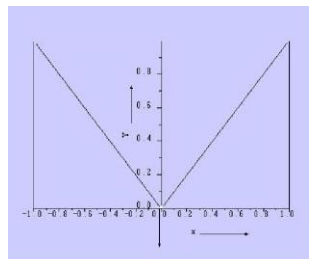
Even and Odd functions:

Definition: A function $y = f(x)$ is said to be even, if $f(-x) = f(x)$. The graph of the even function is always symmetrical about the y-axis.

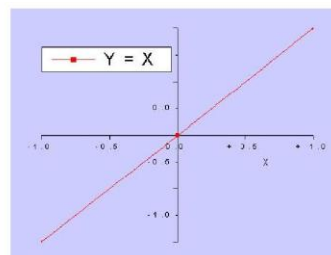
A function $y = f(x)$ is said to be odd, if $f(-x) = -f(x)$. The graph of the odd function is always symmetrical about the origin.

Examples: $f(x) = |x|$ is an even function.

$f(x) = x$ is an odd function.



Graph of $f(x) = |x|$



Graph of $f(x) = x$

Note that the graph of $f(x) = |x|$ is symmetrical about the y-axis and the graph of $f(x) = x$ is symmetrical about the origin.

Note:

1. If $f(x)$ is even and $g(x)$ is odd, then

- $h(x) = f(x) g(x)$ is odd
- $h(x) = f(x) f(x)$ is even
- $h(x) = g(x) g(x)$ is even

Examples:

1. $h(x) = x^2 \cos x$ is even, since both x^2 and $\cos x$ are even functions.

2. $h(x) = x \sin x$ is even, since x and $\sin x$ are odd functions.

3. $h(x) = x^2 \sin x$ is odd, since x^2 is even and $\sin x$ is odd.

Dirichlet's conditions:

Consider a real-valued function $f(x)$ which obeys the following conditions called Dirichlet's conditions :

1. $f(x)$ is defined in an interval $(a, a+2l)$, and $f(x+2l) = f(x)$ so that $f(x)$ is a periodic function of period $2l$.
2. $f(x)$ is continuous or has only a finite number of discontinuities in the interval $(a, a+2l)$.
3. $f(x)$ has no or only a finite number of maxima or minima in the interval $(a, a+2l)$.

Here

$$a_0 = \frac{1}{l} \int_a^{a+2l} f(x) dx, \quad (1)$$

$$a_n = \frac{1}{l} \int_a^{a+2l} f(x) \cos\left(\frac{n\pi}{l}x\right) dx, \quad n=1,2,3,\dots \quad (2)$$

$$b_n = \frac{1}{l} \int_a^{a+2l} f(x) \sin\left(\frac{n\pi}{l}x\right) dx, \quad n=1,2,3,\dots \quad (3)$$

Then the infinite series

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi}{l}x\right) + b_n \sin\left(\frac{n\pi}{l}x\right), \quad (4)$$

is called the Fourier series of $f(x)$ in the interval $(a, a+2l)$. Also, the real numbers $a_0, a_1, a_2, \dots, a_n$, and $b_1, b_2, \dots, b_n$ are called the Fourier coefficients of $f(x)$. The formulae (1), (2) and (3) are called **Euler's formulae**.

It can be proved that the sum of the series (4) is $f(x)$ if $f(x)$ is continuous at x . Thus

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi}{l}x\right) + \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi}{l}x\right). \quad (5)$$

Suppose $f(x)$ is discontinuous at x , then the sum of the series (4) would be

$$\frac{1}{2} [f(x^+) + f(x^-)],$$

where $f(x^+)$ and $f(x^-)$ are the values of $f(x)$ immediately to the right and to the left of $f(x)$ respectively.

Particular Cases:

Case (i)

Suppose $a = 0$. Then $f(x)$ is defined over the interval $(0, 2l)$. Formulae (1), (2), (3) reduce to

$$\begin{aligned} a_0 &= \frac{1}{l} \int_0^{2l} f(x) dx, \\ a_n &= \frac{1}{l} \int_0^{2l} f(x) \cos\left(\frac{n\pi}{l}x\right) dx, \\ b_n &= \frac{1}{l} \int_0^{2l} f(x) \sin\left(\frac{n\pi}{l}x\right) dx, \quad n = 1, 2, \dots, \infty, \end{aligned} \quad (6)$$

Then the right-hand side of (5) is the Fourier expansion of $f(x)$ over the interval $(0, 2l)$.

If $l = \pi$, then $f(x)$ is defined over the interval $(0, 2\pi)$. Formulae (6) reduces to

$$\begin{aligned} a_0 &= \frac{1}{\pi} \int_0^{2\pi} f(x) dx, \\ a_n &= \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx dx, \quad n = 1, 2, \dots, \infty, \end{aligned} \quad (7)$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx dx, \quad n = 1, 2, \dots, \infty.$$

Also, in this case, (5) becomes

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx. \quad (8)$$

Case (ii)

Suppose $a = -l$. Then $f(x)$ is defined over the interval $(-l, l)$. Formulae (1), (2), (3) reduce to

$$\begin{aligned} a_0 &= \frac{1}{l} \int_{-l}^l f(x) dx, \\ a_n &= \frac{1}{l} \int_{-l}^l f(x) \cos\left(\frac{n\pi}{l}x\right) dx, \\ b_n &= \frac{1}{l} \int_{-l}^l f(x) \sin\left(\frac{n\pi}{l}x\right) dx. \end{aligned} \quad (9)$$

Then the right-hand side of (5) is the Fourier expansion of $f(x)$ over the interval $(-l, l)$.

If $l = \pi$, then $f(x)$ is defined over the interval $(-\pi, \pi)$. Formulae (9) reduce to

$$\begin{aligned} a_0 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx, \\ a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx, \quad n = 1, 2, \dots, \infty \end{aligned} \quad (10)$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx, \quad n = 1, 2, \dots, \infty.$$

Putting $l = \pi$ in (5),

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + b_n \sin nx.$$

Expressing $\cos nx$ and $\sin nx$ in exponential form, complex form of Fourier series is obtained.

Complex form of Fourier Series:

The Complex Fourier Series of $f(x)$ is defined to be

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx/l}, \text{ where } c_n = \frac{1}{2l} \int_a^{a+2l} f(x) e^{-inx/l} dx.$$

In particular when $a = -l$, $l = \pi$,

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx}, \text{ where } c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx.$$

Here $c_n = \frac{1}{2}(a_n - ib_n)$, $c_{-n} = \frac{1}{2}(a_n + ib_n)$ and $c_0 = \frac{1}{2}a_0$, $n > 0$,

$$\Rightarrow a_0 = 2c_0, \quad a_n = c_n + c_{-n}, \quad b_n = i(c_n - c_{-n}).$$

Examples:

1. Find the complex Fourier series of $f(x) = x$, $-\pi < x < \pi$. Hence deduce the trigonometric series.

Solution:

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{inx}, \text{ where } c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-inx} dx$$

$$c_n = \frac{1}{2\pi} \int_{-\pi}^{\pi} x e^{-inx} dx$$

$$= -\frac{(-1)^n}{in}, n \neq 0$$

$$\text{for } n = 0, c_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} x dx = 0$$

$$\text{Thus, } f(x) = \sum_{n=-\infty, n \neq 0}^{\infty} \frac{(-1)^{n+1}}{in} e^{inx}$$

Corresponding trigonometric series:

$$a_0 = 2c_0 = 0, \quad a_n = c_n + c_{-n} = 0, \quad b_n = i(c_n - c_{-n}) = 2 \frac{(-1)^{n+1}}{n},$$

Required trigonometric series is

$$f(x) = 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin nx.$$

2. Obtain the complex Fourier series for the function $f(x) = \begin{cases} 0, & \text{for } -\pi < x < 0 \\ \sin x, & \text{for } 0 < x < \pi \end{cases}$.

Hence deduce the trigonometric series.

Solution: Given $f(x) = \begin{cases} 0, & \text{for } -\pi < x < 0 \\ \sin x, & \text{for } 0 < x < \pi \end{cases}$.

The complex form of Fourier series is given by:

$$\begin{aligned} f(x) &= \sum_{n=-\infty}^{\infty} c_n e^{inx} \\ \text{where } c_n &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{inx} dx, \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) \sin x dx, \\ &= \frac{1}{2\pi} \left[\frac{e^{-inx}}{(-in)^2 + 1^2} \{-in \sin x - 1 \cdot \cos x\} \right]_0^{\pi}, \\ &= \frac{1}{2\pi} \left[\frac{e^{-inx}}{1-n^2} \{-in \sin x - 1 \cdot \cos x\} \right]_0^{\pi}, \quad \text{for } n \neq \pm 1 \\ &= \frac{1}{2\pi(1-n^2)} [-\{e^{-inx} \cos \pi - e^0 \cos 0\}], \\ c_n &= \frac{1}{2\pi(1-n^2)} [(-1)^n + 1], \\ &= -\frac{[(-1)^n + 1]}{2\pi(n^2 - 1)}. \end{aligned}$$

$$\text{For } n=1, \quad c_1 = \frac{1}{2\pi} \int_0^{\pi} e^{-inx} \sin x dx = \frac{-i}{4}.$$

$$\text{For } n=-1, \quad c_{-1} = \frac{1}{2\pi} \int_0^{\pi} e^{ix} \sin x dx = \frac{i}{4}.$$

Substituting the above values in (1)

$$f(x) = c_0 + c_1 e^{ix} + \sum_{n=2}^{\infty} c_n e^{inx} + c_{-1} e^{-ix} + \sum_{n=-\infty}^{-2} c_n e^{inx},$$

$$f(x) = \frac{1}{\pi} - \frac{i}{4} e^{ix} + \frac{i}{4} e^{-ix} - \frac{1}{2\pi} \left\{ \sum_{n=2}^{\infty} \frac{1+(-1)^n}{n^2-1} e^{inx} + \sum_{n=-\infty}^{-2} \frac{1+(-1)^n}{n^2-1} e^{inx} \right\}.$$

Corresponding trigonometric series:

$$a_0 = 2c_0 = \frac{2}{\pi}, \quad a_n = c_n + c_{-n} = -\frac{[1+(-1)^n]}{\pi(n^2-1)}, \quad n \geq 2, \quad a_1 = c_1 + c_{-1} = 0 \quad b_n = i(c_n - c_{-n}) = 0, \quad n \geq 2,$$

$$b_1 = i(c_1 - c_{-1}) = \frac{1}{2},$$

$$f(x) = \frac{1}{\pi} + \frac{\sin x}{2} - \frac{2}{\pi} \sum_{m=1}^{\infty} \frac{\cos 2mx}{4m^2 - 1}.$$

3. Find the complex form of the Fourier Series of $f(x) = e^x$ in $-1 \leq x \leq 1$. Hence deduce the trigonometric series.

Solution: The complex form of Fourier series is given by

$$\begin{aligned} f(x) &= \sum_{n=-\infty}^{\infty} c_n e^{inx}, \quad (l=1) \\ \text{while } c_n &= \frac{1}{2} \int_{-1}^1 e^{-x} e^{in\pi x} dx, \\ &= \frac{1}{2} \int_{-1}^1 e^{-(1+in\pi)x} dx, \\ &= \frac{1}{2} \left[\frac{e^{-(1+in\pi)x}}{-(1+in\pi)} \right]_{-1}^1, \\ &= \frac{1}{2} \left[\frac{e^{1+in\pi} - e^{-1-in\pi}}{(1+in\pi)} \cdot \frac{1-in\pi}{1+in\pi} \right], \end{aligned}$$

$$e^{-x} = \sum_{n=-\infty}^{\infty} \frac{(-1)^n (1 - in\pi)}{1^2 + n^2 \pi^2} \sinh 1 e^{in\pi x}.$$

Exercise:

Obtain the Complex Fourier Series expansion of the following functions:

1. $f(x) = \begin{cases} 0, & 0 < x < l \\ a, & 0 < x < 2l \end{cases}$ over the interval $(0, 2l)$. Hence deduce the trigonometric series.

Answer: $f(x) = \frac{a}{2} - \frac{ia}{2\pi} \sum_{n=-\infty}^{\infty} \frac{[1 - (-1)^n]}{n} e^{i\left(\frac{n\pi}{l}\right)x}, n \neq 0.$

$$f(x) = \frac{a}{2} - \frac{a}{\pi} \sum_{n=1}^{\infty} \frac{[1 - (-1)^n]}{n} \sin\left(\frac{n\pi}{l}\right)x, \quad n \neq 0.$$

2. $f(x) = x \sin x$ over the interval $(0, 2\pi)$. Hence deduce the corresponding trigonometric series.

Answer: $f(x) = -1 - \left(\frac{1}{4} + \frac{\pi}{2}i\right)e^{ix} - \left(\frac{1}{4} - \frac{\pi}{2}i\right)e^{-ix} + \sum_{n=-\infty}^{-2} \frac{1}{n^2 - 1} e^{inx} + \sum_{n=2}^{\infty} \frac{1}{n^2 - 1} e^{inx},$

$$f(x) = -1 - \frac{1}{2} \cos x + \pi \sin x + 2 \sum_{n=2}^{\infty} \frac{1}{n^2 - 1} \cos nx.$$

3. $f(x) = \begin{cases} -k, & -\pi < x < 0 \\ k, & 0 < x < \pi \end{cases}$ where 'k' is a real constant over the interval $(-\pi, \pi)$.

Answer: $f(x) = -\frac{ik}{\pi} \sum_{n=-\infty}^{\infty} \frac{[1 - (-1)^n]}{n} e^{inx}, n \neq 0,$

$$f(x) = \frac{2k}{\pi} \sum_{n=1}^{\infty} \frac{[1 - (-1)^n]}{n} \sin nx.$$

Examples on time periodic signals:

Special wave forms:

Special wave forms can be classified as:

1. Square wave
2. Half wave rectifier
3. Saw tooth rectifier
4. Triangular wave

Saw tooth wave form:

1. Express $f(x) = x/2$ as a Fourier series in the interval $-\pi < x < \pi$.

Solution: Since $f(-x) = -x/2 = -f(x)$.

Therefore, $f(x)$ is an odd function and

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx,$$

$$b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx \, dx,$$

$$b_n = \frac{2}{\pi} \int_0^{\pi} \frac{x}{2} \sin nx \, dx,$$

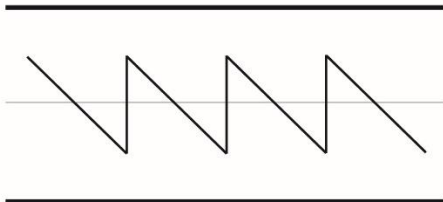
$$b_n = \frac{1}{\pi} \left[x \left(-\frac{\cos nx}{n} \right) - \left(-\frac{\sin nx}{n^2} \right) \right] = \frac{\cos n\pi}{n},$$

$$b_1 = 1, b_2 = -\frac{1}{2}, b_3 = \frac{1}{3}, b_4 = -\frac{1}{4} \text{ etc.....}$$

$$\text{Hence the series is } \frac{x}{2} = \sin x - \frac{1}{2} \sin 2x + \frac{1}{3} \sin 3x - \frac{1}{4} \sin 4x + \dots$$

The above series represents the discontinuous function called saw-tooth waveform shown in the figure below.

SAWTOOTH WAVE



Triangular waveform:

$$2. \text{ Obtain Fourier series for the function } f(x) \text{ given by } f(x) = \begin{cases} 1 + \frac{2x}{\pi}, & -\pi \leq x \leq 0 \\ 1 - \frac{2x}{\pi}, & 0 \leq x \leq \pi \end{cases}$$

Solution: Since $f(-x) = 1 - \frac{2x}{\pi}$ in $(-\pi, 0) = f(x)$ in $(0, \pi)$,

$$f(x) = 1 + \frac{2x}{\pi} \text{ in } (0, \pi) = f(x) \text{ in } (-\pi, 0),$$

therefore $f(x)$ is an even function in $(-\pi, \pi)$ and the graph is symmetrical about the y-axis, the required series is

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx),$$

$$\text{where, } a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx = \frac{2}{\pi} \int_0^{\pi} \left(1 - \frac{2x}{\pi} \right) dx = 0,$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx = \frac{2}{\pi} \int_0^{\pi} \left(1 - \frac{2x}{\pi} \right) \cos nx dx,$$

$$= \frac{4}{n^2 \pi^2} [1 - (-1)^n],$$

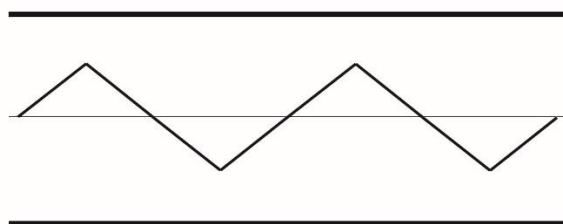
$$a_1 = \frac{8}{\pi^2}, a_3 = \frac{8}{3^2 \pi^2}, a_5 = \frac{8}{5^2 \pi^2}, \dots$$

$$a_2 = a_4 = a_6 = \dots = 0.$$

Thus substituting the values of a 's in the required series,

$$f(x) = \frac{8}{\pi^2} \left[\frac{\cos x}{1^2} + \frac{\cos 3x}{3^2} + \frac{\cos 5x}{5^2} + \dots \right] \text{ is the required Fourier series.}$$

TRIANGLE WAVE



Square wave form:

3. Find the Fourier series of the periodic function $f(x) = \begin{cases} -k, & -\pi < x < 0 \\ k, & 0 < x < \pi \end{cases}$.

Functions of this kind occur as external forces acting on mechanical systems, electromotive forces in electric circuits, etc.

Solution: The Fourier series of $f(x)$ defined in $(-\pi, \pi)$ having a period of 2π is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx \dots,$$

$$a_0 = \frac{1}{\pi} \left\{ \int_{-\pi}^0 -k dx + \int_0^{\pi} k dx \right\},$$

$$= \frac{k}{\pi} \{0\} = 0,$$

$$\frac{a_0}{2} = 0,$$

$$a_n = \frac{1}{\pi} \left\{ \int_{-\pi}^0 -k \cos nx dx + \int_0^{\pi} k \cos nx dx \right\},$$

$$a_n = 0,$$

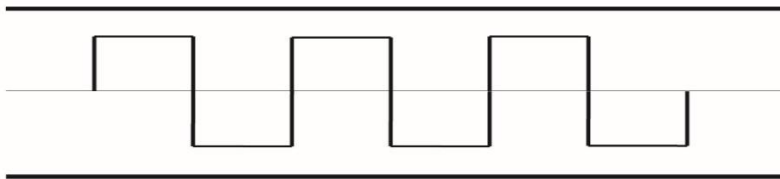
$$b_n = \frac{1}{\pi} \left\{ \int_{-\pi}^0 -k \sin nx dx + \int_0^{\pi} k \sin nx dx \right\},$$

$$b_n = \frac{2k}{\pi n} \{1 - (-1)^n\}.$$

The required Fourier series is given by

$$f(x) = \sum_{n=1}^{\infty} \frac{2k}{\pi n} \{1 - (-1)^n\} \sin nx.$$

SQUARE WAVE



4. An alternating current after passing through a rectifier has the form

$$I = \begin{cases} -I_0 \sin \theta, & 0 < \theta < \pi \\ 0, & \pi < \theta < 2\pi \end{cases} \quad \text{where } I_0 \text{ is the max current.}$$

Express I as Fourier series in $(0, 2\pi)$.

Solution: The Fourier series of period 2π is given by

$$I = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos n\theta + \sum_{n=1}^{\infty} b_n \sin n\theta,$$

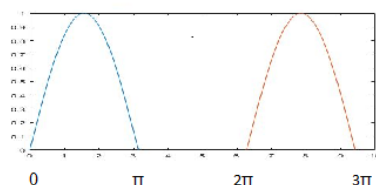
$$a_0 = \frac{I_0}{\pi} \int_0^{\pi} \sin \theta d\theta, \quad a_n = \frac{I_0}{\pi} \int_0^{\pi} \sin \theta \cos n\theta d\theta, \quad b_n = \frac{I_0}{\pi} \int_0^{\pi} \sin \theta \sin n\theta d\theta,$$

$$a_0 = \frac{2I_0}{\pi},$$

$$a_n = \frac{-I_0}{\pi(n^2-1)} \{1 + (-1)^n\} \text{ if } n \neq 1, \quad a_1 = 0, \quad b_n = 0 \text{ if } n \neq 1 \text{ and } b_1 = \frac{I_0}{2}.$$

Substituting in (1);

$$I = \frac{I_0}{\pi} + \sum_{n=2}^{\infty} \frac{-1}{\pi(n^2-1)} \{1 + (-1)^n\} \cos n\theta + \frac{I_0}{2} \sin \theta.$$



Example for Fourier expansion of continuous functions:

1. Obtain the Fourier expansion of $f(x) = \frac{1}{2}(\pi - x)$ in $-\pi < x < \pi$.

Solution:

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_{-\pi}^{\pi} \frac{1}{2}(\pi - x) dx,$$

$$= \frac{1}{2\pi} \left[\pi x - \frac{x^2}{2} \right]_{-\pi}^{\pi} = \pi,$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{1}{\pi} \int_{-\pi}^{\pi} \frac{1}{2}(\pi - x) \cos nx dx,$$

Integration by parts,

$$a_n = \frac{1}{2\pi} \left[(\pi - x) \frac{\sin nx}{n} - (-1) \left(\frac{-\cos nx}{n^2} \right) \right]_{-\pi}^{\pi},$$

$$= \frac{1}{2\pi} [0] = 0,$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} \frac{1}{2}(\pi - x) \sin nx dx,$$

$$= \frac{1}{2\pi} \left[(\pi - x) \frac{-\cos nx}{n} - (-1) \left(\frac{-\sin nx}{n^2} \right) \right]_{-\pi}^{\pi},$$

$$= \frac{(-1)^n}{n}.$$

Using the values of a_0 , a_n and b_n in the Fourier expansion

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx + \sum_{n=1}^{\infty} b_n \sin nx,$$

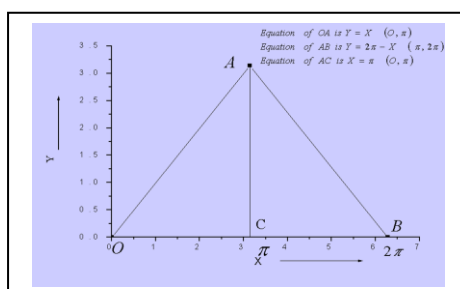
$$f(x) = \frac{\pi}{2} + \sum_{n=1}^{\infty} \frac{(-1)^n}{n} \sin nx.$$

This is the required Fourier expansion of the given function.

2. Obtain the Fourier expansion of

$$f(x) = \begin{cases} x, & 0 \leq x \leq \pi \\ 2\pi - x, & \pi \leq x \leq 2\pi \end{cases}. \text{ Deduce that } \frac{\pi^2}{8} = 1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots$$

Solution: The graph of $f(x)$ is shown below.



Here OA represents the line $f(x) = x$, AB represents the line $f(x) = (2\pi - x)$ and AC represents the line $x = \pi$. The graph is symmetrical about the line AC, which in turn is parallel to y-axis. Hence the function $f(x)$ is an even function.

$$\text{Here, } a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{2}{\pi} \int_0^{\pi} f(x) dx,$$

$$= \frac{2}{\pi} \int_0^{\pi} x dx = \pi$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx$$

Since $f(x) \cos nx$ is even.

$$= \frac{2}{\pi} \int_0^{\pi} x \cos nx dx = \frac{2}{\pi} \left[x \left(\frac{\sin nx}{n} \right) - 1 \left(\frac{-\cos nx}{n^2} \right) \right]_0^{\pi} = \frac{2}{n^2 \pi} [(-1)^n - 1].$$

$$\text{Also, } b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx = 0, \text{ since } f(x) \sin nx \text{ is odd.}$$

$$\text{Thus the Fourier series of } f(x) \text{ is } f(x) = \frac{\pi}{2} + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1}{n^2} [(-1)^n - 1] \cos nx.$$

For $x = \pi$ a point of continuity,

$$f(\pi) = \frac{\pi}{2} + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1}{n^2} [(-1)^n - 1] \cos n\pi,$$

$$\text{or } \pi = \frac{\pi}{2} + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{-2 \cos(2n-1)\pi}{(2n-1)^2}.$$

$$\text{Thus, } \frac{\pi^2}{8} = \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2} \text{ or } \frac{\pi^2}{8} = 1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots.$$

This is the series as required.

Example for discontinuous functions:

3. Obtain the Fourier expansion of

$$f(x) = \begin{cases} -\pi, & -\pi < x < 0 \\ x, & 0 < x < \pi. \end{cases}$$

$$\text{Deduce that } \frac{\pi^2}{8} = 1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots.$$

$$\begin{aligned} \text{Solution: Here, } a_0 &= \frac{1}{\pi} \left[\int_{-\pi}^0 -\pi dx + \int_0^{\pi} x dx \right] = -\frac{\pi}{2}, \\ a_n &= \frac{1}{\pi} \left[\int_{-\pi}^0 -\pi \cos nx dx + \int_0^{\pi} x \cos nx dx \right], \\ &= \frac{1}{n^2 \pi} [(-1)^n - 1], \\ b_n &= \frac{1}{\pi} \left[\int_{-\pi}^0 -\pi \sin nx dx + \int_0^{\pi} x \sin nx dx \right], \\ &= \frac{1}{n} [1 - 2(-1)^n]. \end{aligned}$$

Fourier series is $f(x) = \frac{-\pi}{4} - \frac{1}{\pi} \sum_{n=1}^{\infty} \frac{1}{n^2} [(-1)^n - 1] \cos nx + \sum_{n=1}^{\infty} \frac{[1 - 2(-1)^n]}{n} \sin nx$.

Note, that the point $x = 0$ is a point of discontinuity of $f(x)$.

Here $f(x^+) = 0$, $f(x^-) = -\pi$ at $x = 0$. Hence $\frac{1}{2} [f(x^+) + f(x^-)] = \frac{1}{2} (0 - \pi) = \frac{-\pi}{2}$.

4. Obtain the Fourier expansion of $f(x) = e^{-ax}$ in the interval $(-\pi, \pi)$. Deduce that

$$\pi \operatorname{cosech} \pi = 2 \sum_{n=2}^{\infty} \frac{(-1)^n}{n^2 + 1}.$$

Solution: Here,

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} e^{-ax} dx = \frac{1}{\pi} \left[\frac{e^{-ax}}{-a} \right]_{-\pi}^{\pi} = \frac{e^{a\pi} - e^{-a\pi}}{a\pi} = \frac{2 \sinh a\pi}{a\pi},$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} e^{-ax} \cos nx dx$$

$$a_n = \frac{1}{\pi} \left[\frac{e^{-ax}}{a^2 + n^2} \{-a \cos nx + n \sin nx\} \right]_{-\pi}^{\pi} = \frac{2a}{\pi} \left[\frac{(-1)^n \sinh a\pi}{a^2 + n^2} \right]$$

$$b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} e^{-ax} \sin nx dx$$

$$= \frac{1}{\pi} \left[\frac{e^{-ax}}{a^2 + n^2} \{-a \sin nx - n \cos nx\} \right]_{-\pi}^{\pi}$$

$$= \frac{2n}{\pi} \left[\frac{(-1)^n \sinh a\pi}{a^2 + n^2} \right].$$

Thus, $f(x) = \frac{\sinh a\pi}{a\pi} + \frac{2a \sinh a\pi}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{a^2 + n^2} \cos nx + \frac{2}{\pi} \sinh a\pi \sum_{n=1}^{\infty} \frac{n(-1)^n}{a^2 + n^2} \sin nx$.

For $x = 0$, $a = 1$, the series reduces to

$$f(0) = 1 = \frac{\sinh \pi}{\pi} + \frac{2 \sinh \pi}{\pi} \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2 + 1}$$

or

$$1 = \frac{\sinh \pi}{\pi} + \frac{2 \sinh \pi}{\pi} \left[-\frac{1}{2} + \sum_{n=2}^{\infty} \frac{(-1)^n}{n^2 + 1} \right]$$

or

$$1 = \frac{2 \sinh \pi}{\pi} \sum_{n=2}^{\infty} \frac{(-1)^n}{n^2 + 1}$$

Thus,

$$\pi \operatorname{cosech} \pi = 2 \sum_{n=2}^{\infty} \frac{(-1)^n}{n^2 + 1}.$$

This is the desired deduction.

Example for even and odd functions:

5. Obtain the Fourier expansion of $f(x) = x^2$ over the interval $(-\pi, \pi)$. Deduce that

$$\frac{\pi^2}{6} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots + \infty.$$

Solution: The function $f(x)$ is even. Hence

$$a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{2}{\pi} \int_0^{\pi} f(x) dx,$$

$$= \frac{2}{\pi} \int_0^{\pi} x^2 dx = \frac{2}{\pi} \left[\frac{x^3}{3} \right]_0^{\pi},$$

$$a_0 = \frac{2\pi^2}{3}.$$

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx,$$

$$= \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx, \quad \text{since } f(x) \cos nx \text{ is even}$$

$$= \frac{2}{\pi} \int_0^{\pi} x^2 \cos nx dx,$$

Integrating by parts,

$$a_n = \frac{2}{\pi} \left[x^2 \left(\frac{\sin nx}{n} \right) - 2x \left(\frac{-\cos nx}{n^2} \right) + 2 \left(\frac{-\sin nx}{n^3} \right) \right]_0^{\pi} = \frac{4(-1)^n}{n^2}.$$

$$\text{Also, } b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx = 0, \quad \text{since } f(x) \sin nx \text{ is odd.}$$

$$f(x) = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n \cos nx}{n^2}, \quad (a)$$

$$\pi^2 = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{1}{n^2}, \quad (\text{putting } x = \pi \text{ in (a)})$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}.$$

$$\text{Hence, } \frac{\pi^2}{6} = 1 + \frac{1}{2^2} + \frac{1}{3^2} + \dots.$$

Example for arbitrary interval

6. Obtain the Fourier series of $f(x) = 1 - x^2$ over the interval $(-1, 1)$.

Solution: The given function is even as $f(-x) = f(x)$. Also period of $f(x)$ is $1 - (-1) = 2$.

$$\text{Here } a_0 = \frac{1}{1} \int_{-1}^1 f(x) dx = 2 \int_0^1 f(x) dx,$$

$$\begin{aligned}
 &= 2 \int_0^1 (1-x^2) dx = 2 \left[x - \frac{x^3}{3} \right]_0^1, \\
 &= \frac{4}{3}, \\
 a_n &= \frac{1}{1} \int_{-1}^1 f(x) \cos(n\pi x) dx, \\
 &= 2 \int_0^1 f(x) \cos(n\pi x) dx, \quad \text{as } f(x) \cos(n\pi x) \text{ is even} \\
 &= 2 \int_0^1 (1-x^2) \cos(n\pi x) dx.
 \end{aligned}$$

Integrating by parts,

$$\begin{aligned}
 a_n &= 2 \left[(1-x^2) \left(\frac{\sin n\pi x}{n\pi} \right) - (-2x) \left(\frac{-\cos n\pi x}{(n\pi)^2} \right) + (-2) \left(\frac{-\sin n\pi x}{(n\pi)^3} \right) \right]_0^1, \\
 &= \frac{4(-1)^{n+1}}{n^2 \pi^2}, \\
 b_n &= \frac{1}{1} \int_{-1}^1 f(x) \sin(n\pi x) dx = 0, \quad \text{since } f(x) \sin(n\pi x) \text{ is odd.}
 \end{aligned}$$

The Fourier series of $f(x)$ is

$$f(x) = \frac{2}{3} + \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2} \cos(n\pi x)$$

7. Obtain the Fourier expansion of

$$f(x) = \begin{cases} 1 + \frac{4x}{3} & \text{in } -\frac{3}{2} < x \leq 0 \\ 1 - \frac{4x}{3} & \text{in } 0 \leq x < \frac{3}{2} \end{cases}.$$

Deduce that $\frac{\pi^2}{8} = 1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots$

Solution: The period of $f(x)$ is $\frac{3}{2} - \left(-\frac{3}{2}\right) = 3$.

Also $f(-x) = f(x)$. Hence $f(x)$ is even.

$$\begin{aligned}
 a_0 &= \frac{1}{3/2} \int_{-3/2}^{3/2} f(x) dx = \frac{2}{3/2} \int_0^{3/2} f(x) dx, \\
 &= \frac{4}{3} \int_0^{3/2} \left(1 - \frac{4x}{3}\right) dx = 0,
 \end{aligned}$$

$$\begin{aligned}
 a_n &= \frac{1}{3/2} \int_{-3/2}^{3/2} f(x) \cos\left(\frac{n\pi x}{3/2}\right) dx, \\
 &= \frac{2}{3/2} \int_0^{3/2} f(x) \cos\left(\frac{2n\pi x}{3}\right) dx,
 \end{aligned}$$

$$= \frac{4}{3} \left(1 - \frac{4x}{3} \right) \left(\frac{\sin \left(\frac{2n\pi x}{3} \right)}{\left(\frac{2n\pi}{3} \right)} \right) - \left(\frac{-4}{3} \right) \left(\frac{-\cos \left(\frac{2n\pi x}{3} \right)}{\left(\frac{2n\pi}{3} \right)^2} \right) \Bigg|_0^{3/2}$$

$$= \frac{4}{n^2 \pi^2} [1 - (-1)^n].$$

Also,
$$b_n = \frac{1}{3} \int_{-3/2}^{3/2} f(x) \sin \left(\frac{n\pi x}{3/2} \right) dx = 0.$$

Thus
$$f(x) = \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} [1 - (-1)^n] \cos \left(\frac{2n\pi x}{3} \right).$$

Putting $x = 0$,
$$f(0) = \frac{4}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} [1 - (-1)^n],$$

or $1 = \frac{8}{\pi^2} \left[1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots \right]$. Thus, $\frac{\pi^2}{8} = 1 + \frac{1}{3^2} + \frac{1}{5^2} + \dots$

Exercise:

Obtain the Fourier series of the following functions over the specified intervals :

1. $f(x) = x + x^2$ over $(-\pi, \pi)$.

Answer:

$$f(x) = \frac{\pi^2}{3} + 4 \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos nx + 2 \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} \sin nx.$$

2. $f(x) = \left(\frac{\pi - x}{2} \right)^2$ over $(0, 2\pi)$.

Answer:

$$f(x) = \frac{\pi^2}{6} + \sum_{n=1}^{\infty} \frac{1}{n^2} \cos nx.$$

3. $f(x) = x$ over $(-\pi, \pi)$; Deduce that $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \dots \infty$.

Answer:

$$f(x) = 2 \sum_{n=1}^{\infty} \frac{1}{n} (-1)^{n+1} \sin nx.$$

Deduction at $x = \frac{\pi}{2}$.

4. $f(x) = |x|$ over $(-\pi, \pi)$; Deduce that $\frac{\pi^2}{8} = \frac{1}{1^2} + \frac{1}{3^2} + \dots \infty$.

Answer:

$$f(x) = \frac{\pi}{2} + 2 \sum_{n=1}^{\infty} \frac{1}{\pi n^2} [(-1)^n - 1] \cos nx.$$

Deduction at $x = \pi$.

5. $f(x) = x \sin x$ over $0 \leq x \leq 2\pi$. Deduce that

$$\sum_{n=2}^{\infty} \frac{1}{n^2 - 1} = \frac{3}{4}.$$

6. $f(x) = \begin{cases} 0, & -2 \leq x \leq 0 \\ 1, & 0 < x \leq 2 \end{cases}$ over $(-2, 2)$.

Answer:

$$f(x) = \frac{1}{2} + \frac{2}{\pi} \sum_{n=1}^{\infty} \frac{1}{(2n-1)} \sin \left[\frac{(2n-1)\pi x}{2} \right].$$

7. $f(x) = x(2-x)$ over $(0,3)$.

Answer:

$$f(x) = -\frac{9}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} \cos \left(\frac{2n\pi x}{3} \right) + \frac{3}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} \sin \left(\frac{2n\pi x}{3} \right).$$

8. $f(x) = x$ over $(-l, l)$.

Answer:

$$f(x) = \frac{2l}{\pi} \sum_{n=1}^{\infty} \frac{1}{n} (-1)^{n+1} \sin \left(\frac{n\pi x}{l} \right).$$

9. $f(x) = \begin{cases} \pi x, & 0 \leq x \leq 1 \\ \pi(2-x), & 1 \leq x \leq 2. \end{cases}$ over the interval $(0,2)$. Deduce that $\frac{\pi^2}{8} = \sum_{n=1}^{\infty} \frac{1}{(2n-1)^2}$.

Answer:

$$f(x) = \frac{\pi}{2} + 2 \sum_{n=1}^{\infty} \frac{1}{\pi n^2} [(-1)^n - 1] \cos nx.$$

Deduction at $x = 1$.

Half-range Fourier series:

The Fourier expansion of the periodic function $f(x)$ of period $2l$ may contain both sine and cosine terms. If it is required to obtain the Fourier expansion of $f(x)$ in the interval $(0, l)$ which is regarded as half interval. The definition can be extended to the other half in such a manner that the function becomes even or odd. This will result in cosine series or sine series only.

Sine series:

Suppose $f(x) = \phi(x)$ is given in the interval $(0, l)$. Then defining $f(x) = -\phi(-x)$ in $(-l, 0)$. Hence $f(x)$ becomes an odd function in $(-l, l)$. The Fourier series then is

$$f(x) = \sum_{n=1}^{\infty} b_n \sin \left(\frac{n\pi x}{l} \right), \quad (11)$$

where $b_n = \frac{2}{l} \int_0^l f(x) \sin \left(\frac{n\pi x}{l} \right) dx.$

The series (11) is called half-range sine series over $(0, l)$.

Putting $l = \pi$ in (11), the half-range sine series of $f(x)$ over $(0, \pi)$ is obtained, given by

$$f(x) = \sum_{n=1}^{\infty} b_n \sin nx \quad b_n = \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx dx.$$

Cosine series: Let us define

$$f(x) = \begin{cases} \phi(x) & \text{in } (0, l) \quad \dots \text{given} \\ \phi(-x) & \text{in } (-l, 0) \quad \dots \text{in order to make the function even.} \end{cases}$$

Then the Fourier series of $f(x)$ is given by

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi x}{l}\right), \quad (12)$$

$$a_0 = \frac{2}{l} \int_0^l f(x) dx,$$

$$a_n = \frac{2}{l} \int_0^l f(x) \cos\left(\frac{n\pi x}{l}\right) dx.$$

The series (12) is called half-range cosine series over $(0, l)$.

Putting $l = \pi$ in (12),

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos nx,$$

where

$$a_0 = \frac{2}{\pi} \int_0^{\pi} f(x) dx,$$

$$a_n = \frac{2}{\pi} \int_0^{\pi} f(x) \cos nx dx \quad n = 1, 2, 3, \dots$$

1. Expand $f(x) = x(\pi - x)$ as half-range sine series over the interval $(0, \pi)$.

Solution:

$$\begin{aligned} b_n &= \frac{2}{\pi} \int_0^{\pi} f(x) \sin nx dx, \\ &= \frac{2}{\pi} \int_0^{\pi} (\pi x - x^2) \sin nx dx, \end{aligned}$$

Integrating by parts,

$$\begin{aligned} b_n &= \frac{2}{\pi} \left[(\pi x - x^2) \left(\frac{-\cos nx}{n} \right) - (\pi - 2x) \left(\frac{-\sin nx}{n^2} \right) + (-2) \left(\frac{\cos nx}{n^3} \right) \right]_0^{\pi}, \\ &= \frac{4}{n^3 \pi} [1 - (-1)^n]. \end{aligned}$$

The sine series of $f(x)$ is

$$f(x) = \frac{4}{\pi} \sum_{n=1}^{\infty} \frac{1}{n^3} [1 - (-1)^n] \sin nx.$$

2. Obtain the cosine series of

$$f(x) = \begin{cases} x, & 0 < x < \frac{\pi}{2} \\ \pi - x, & \frac{\pi}{2} < x < \pi \end{cases} \text{ over } (0, \pi).$$

Solution: Here

$$a_0 = \frac{2}{\pi} \left[\int_0^{\pi/2} x dx + \int_{\pi/2}^{\pi} (\pi - x) dx \right] = \frac{\pi}{2}.$$

$$a_n = \frac{2}{\pi} \left[\int_0^{\pi/2} x \cos nx dx + \int_{\pi/2}^{\pi} (\pi - x) \cos nx dx \right].$$

Performing integration by parts and simplifying,

$$a_n = -\frac{2}{n^2 \pi} \left[1 + (-1)^n - 2 \cos \left(\frac{n\pi}{2} \right) \right],$$

$$= -\frac{8}{n^2 \pi}, \quad n = 2, 6, 10, \dots$$

Thus, the Fourier cosine series is

$$f(x) = \frac{\pi}{4} - \frac{2}{\pi} \left[\frac{\cos 2x}{1^2} + \frac{\cos 6x}{3^2} + \frac{\cos 10x}{5^2} + \dots \infty \right].$$

3. Obtain the half-range cosine series of $f(x) = c - x$ in $0 < x < c$.

Solution: Here

$$a_0 = \frac{2}{c} \int_0^c (c - x) dx = c,$$

$$a_n = \frac{2}{c} \int_0^c (c - x) \cos \left(\frac{n\pi x}{c} \right) dx, \text{ integrating by parts and simplifying,}$$

$$a_n = \frac{2c}{n^2 \pi^2} [1 - (-1)^n].$$

The cosine series is given by

$$f(x) = \frac{c}{2} + \frac{2c}{\pi^2} \sum_{n=1}^{\infty} \frac{1}{n^2} [1 - (-1)^n] \cos \left(\frac{n\pi x}{c} \right).$$

Exercise:

Obtain the half-range sine series of the following functions over the specified intervals:

1. $f(x) = \cos x$ over $(0, \pi)$.

Answer:

$$f(x) = \frac{2}{\pi} \sum_{n=2}^{\infty} \frac{n}{(n^2 - 1)} [1 + (-1)^n] \sin nx.$$

2. The temperature distribution $T(x)$ at a distance x measured from one end along a metal bar of length π is given by $T(x) = x \sin x$ over $(0, \pi)$. Express $T(x)$ as a half range Fourier sine series.

Answer:

$$f(x) = \frac{2}{\pi} \sum_{n=2}^{\infty} \frac{n}{(n^2-1)} [1 + (-1)^n] \sin nx.$$

3. $f(x) = lx - x^2$ over $(0, l)$

Answer:

$$f(x) = \frac{4l^2}{\pi^3} \sum_{n=2}^{\infty} \frac{[1 - (-1)^n]}{n^3} \sin\left(\frac{n\pi}{l}\right)x.$$

Obtain the half-range cosine series of the following functions over the specified intervals :

4. $f(x) = x^2$ over $(0, \pi)$

Answer:

$$f(x) = \frac{\pi^2}{3} - 4 \sum_{n=2}^{\infty} \frac{(-1)^{n+1}}{n^2} \cos nx.$$

5. $f(x) = (x-1)^2$ over $(0, 1)$.

Answer:

$$f(x) = \frac{1}{3} + \frac{4}{\pi^2} \sum_{n=2}^{\infty} \frac{1}{n^2} \cos n\pi x.$$

Obtain the half-range cosine and sine series of the following functions over the specified intervals :

$$6. f(x) = \begin{cases} kx, & 0 \leq x \leq \frac{l}{2} \\ k(l-x), & \frac{l}{2} \leq x \leq l. \end{cases}$$

Answer:

Cosine series:

$$f(x) = \frac{kl}{4} + \frac{2kl}{\pi^2} \sum_{n=2}^{\infty} \frac{\left[2 \cos\left(\frac{n\pi}{2}\right) - 1 - (-1)^n\right]}{n^2} \cos\left(\frac{n\pi}{l}\right)x.$$

Sine series:

$$f(x) = \frac{4kl}{\pi^2} \sum_{n=2}^{\infty} \frac{1}{n^2} \sin\left(\frac{n\pi}{2}\right) \sin\left(\frac{n\pi}{l}\right)x.$$

Harmonic Analysis:

We have discussed the methods of obtaining the Fourier series of a known function $f(x)$ in a given interval. However, there will also be situations where there will be no known functional expression for $f(x)$ but only the values at some equidistant points will be known.

Harmonic Analysis is the process of finding the Fourier series for a function given by numerical values.



Let the Fourier series for $y = f(x)$ in $(0, 2\pi)$ is given as

$$y = \frac{a_0}{2} + a_1 \cos x + a_2 \cos 2x + \dots + b_1 \sin x + b_2 \sin 2x + \dots$$

Where

$$a_0 = \frac{1}{\pi} \int_0^{2\pi} f(x) dx$$

$$a_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \cos nx dx$$

$$b_n = \frac{1}{\pi} \int_0^{2\pi} f(x) \sin nx dx, \quad n=1,2,3,\dots$$

Where $\frac{a_0}{2}$ is called the constant term and the groups of terms $(a_1 \cos x + b_1 \sin x)$, $(a_2 \cos 2x + b_2 \sin 2x)$ etc. are called the first harmonics and second harmonics.

Note: The mean value of a continuous function $y = f(x)$ in the range (a, b) is given by

$$\frac{1}{b-a} \int_a^b f(x) dx$$

Suppose we have a set of N values of $y = f(x)$ having period $2l$ at equidistant points of x in the interval $(c, c + 2l)$. If the values of y at $x = c$ and $x = c + 2l$ are given we must omit one of them. By periodic property $(y)_{x=c} = (y)_{x=c+2l}$. The Fourier coefficients a_0, a_n, b_n assume the following form by the above stated principle.

$$a_0 = \frac{1}{l} \int_c^{c+2l} f(x) dx = 2 \left[\frac{1}{(c+2l)-c} \int_c^{c+2l} f(x) dx \right]$$

$$a_0 = 2 \left[\text{mean value of } f(x) = y \text{ in } (c, c + 2l) \right] = \frac{2}{N} \sum y$$

$$a_n = \frac{1}{l} \int_c^{c+2l} f(x) \cos \left(\frac{n\pi x}{l} \right) dx = 2 \left[\frac{1}{(c+2l)-c} \int_c^{c+2l} f(x) \cos \left(\frac{n\pi x}{l} \right) dx \right]$$

$$a_n = 2 \left[\text{mean value of } f \cos \left(\frac{n\pi x}{l} \right) \text{ in } (c, c + 2l) \right]$$

$$a_n = 2 \left[\frac{\sum y \cos \left(\frac{n\pi x}{l} \right)}{N} \right] = \frac{2}{N} \sum y \cos \left(\frac{n\pi x}{l} \right)$$

$$b_n = 2 \left[\frac{\sum y \sin \left(\frac{n\pi x}{l} \right)}{N} \right] = \frac{2}{N} \sum y \sin \left(\frac{n\pi x}{l} \right)$$

Note: 1. In an interval $(c, c + 2\pi)$, a_n and b_n can be expressed as

$$a_0 = 2 \left[\frac{\sum y}{N} \right] = \frac{2}{N} \sum y$$

$$a_n = 2 \left[\frac{\sum y \cos nx}{N} \right] = \frac{2}{N} \sum y \cos nx$$

$$b_n = 2 \left[\frac{\sum y \sin nx}{N} \right] = \frac{2}{N} \sum y \sin nx$$

2. For Half range cosine Fourier series of $f(x)$ over the interval $(0, l)$, a_0 and a_n are

$$a_0 = 2 \left[\frac{\sum y}{N} \right] = 2 \frac{\sum y}{N}; \quad a_n = 2 \left[\frac{\sum y \cos \left(\frac{n\pi x}{l} \right)}{N} \right] = 2 \frac{\sum y \cos \left(\frac{n\pi x}{l} \right)}{N}$$

3. For Half range sine Fourier series of $f(x)$ over the interval $(0, l)$, b_n is

$$b_n = 2 \left[\frac{\sum y \sin \left(\frac{n\pi x}{l} \right)}{N} \right] = \frac{2}{N} \sum y \sin \left(\frac{n\pi x}{l} \right)$$

Problems:

1. Determine the constant term and the first cosine and sine terms of the Fourier series expansion of y from the following data.

x°	0	45	90	135	180	225	270	315
y	2	1.5	1	0.5	0	0.5	1	1.5

Solution: Here the interval is $(0, 2\pi)$

x°	y	$\cos x$	$y \cos x$	$\sin x$	$y \sin x$
0	2.0	0	2.0	0	0
45	1.5	0.7071	1.06065	0.7071	1.06065
90	1.0	0	0	1	1.0
135	0.5	-0.7071	-0.35355	0.7071	0.35355
180	0	-1	0	0	0
225	0.5	-0.7071	-0.35355	-0.7071	-0.35355
270	1.0	0	0	-1	-1
315	1.5	0.7071	1.06065	-0.7071	-1.06065
Totals	8.0		3.4142		0

$$a_0 = \frac{2}{N} \sum y, \quad a_1 = \frac{2}{N} \sum y \cos x, \quad b_1 = \frac{2}{N} \sum y \sin x$$

From the table $\sum y = 8$, $\sum y \cos x = 3.4142$, $\sum y \sin x = 0$

$$\therefore a_0 = \frac{2}{8}(8.0) = 2, \quad a_1 = \frac{2}{8}(3.4142) = 0.85355, \quad b_1 = \frac{2}{8}(0) = 0$$

Thus, the Fourier series of y up to the first harmonic is

$$y = \frac{a_0}{2} + a_1 \cos x + b_1 \sin x$$

$$\text{i.e. } y = 1 + 0.85355 \cos x$$

2. Analyse harmonically the data given below and express y in Fourier series up to the third harmonic:

x	0	$\frac{\pi}{3}$	$\frac{2\pi}{3}$	π	$\frac{4\pi}{3}$	$\frac{5\pi}{3}$	2π
y	1.0	1.4	1.9	1.7	1.5	1.2	1.0

Solution: Since the last value of y is a repetition of the first, only the first six values will be used. The length of the interval is 2π .

$$\text{Let } y = \frac{a_0}{2} + (a_1 \cos x + b_1 \sin x) + (a_2 \cos 2x + b_2 \sin 2x) + (a_3 \cos 3x + b_3 \sin 3x)$$

x	y	$\cos x$	$\sin x$	$\cos 2x$	$\sin 2x$	$\cos 3x$	$\sin 3x$	$y \cos x$	$y \cos 2x$	$y \cos 3x$	$y \sin x$	$y \sin 2x$	$y \sin 3x$
0	1.0	1	0	1	0	1	0	1	1	1	0	0	0
$\frac{\pi}{3}$	1.4	0.5	0.866	-0.5	0.866	-1	0	0.7	-0.7	-1.4	1.2124	1.2124	0
$\frac{2\pi}{3}$	1.9	-0.5	0.866	-0.5	-0.866	1	0	-0.95	-0.95	1.9	1.6454	-1.6454	0
π	1.7	-1	0	1	0	-1	0	-1.7	1.7	-1.7	0	0	0



$\frac{4\pi}{3}$	1.5	-0.5	0.866	-0.5	0.866	1	0	-0.75	-0.75	1.5	1.299	1.299	0
$\frac{5\pi}{3}$	1.2	0.5	-0.866	-0.5	-0.866	-1	0	0.6	-0.6	-1.2	-1.0392	-1.0392	0
Totals	8.7							-1.1	-0.3	0.1	3.1176	-0.1732	0

$$a_0 = \frac{2}{6}(8.7) = 2.9, \quad a_1 = \frac{2}{6}(-1.1) = -0.37, \quad b_1 = \frac{2}{6}(3.1176) = 1.0392,$$

$$a_2 = \frac{2}{6}(-0.3) = -0.1, \quad b_2 = \frac{2}{6}(-0.1732) = -0.0577, \quad a_3 = \frac{2}{6}(0.1) = 0.0333,$$

$$b_3 = \frac{2}{6}(0) = 0$$

$$\therefore y = 1.45 + (-0.37\cos x + 1.0392\sin x) + (-0.1\cos 2x - 0.0577\sin 2x) + (0.0333\cos 3x)$$

3. Obtain the constant term and the coefficients of the first sine and cosine terms in the Fourier expansion of y as given in the following table:

x	0	1	2	3	4	5
y	9	18	24	28	26	20

Solution: Let the Fourier series to represent y be

$$y = \frac{a_0}{2} + \left(a_1\cos\frac{\pi x}{3} + b_1\sin\frac{\pi x}{3}\right) + \left(a_2\cos\frac{2\pi x}{3} + b_2\sin\frac{2\pi x}{3}\right) + \dots$$

x	y	$\cos\frac{\pi x}{3}$	$\sin\frac{\pi x}{3}$	$y\cos\frac{\pi x}{3}$	$y\sin\frac{\pi x}{3}$
0	9	1	0	9	0
1	18	0.5	$\frac{\sqrt{3}}{2}$	9	$9\sqrt{3}$
2	24	-0.5	$\frac{\sqrt{3}}{2}$	-12	$12\sqrt{3}$
3	28	-1	0	-28	0
4	26	-0.5	$-\frac{\sqrt{3}}{2}$	-13	$-13\sqrt{3}$
5	20	0.5	$-\frac{\sqrt{3}}{2}$	10	$-10\sqrt{3}$
Totals	125			-25	$-2\sqrt{3}$

$$\therefore a_0 = \frac{2}{6}(125) = 41.67, \quad a_1 = \frac{2}{6}(-25) = -8.333, \quad b_1 = \frac{2}{6}(-2\sqrt{3}) = -1.154$$

$$\therefore y = 20.835 + (-8.333\cos x - 1.154\sin x)$$

4. The following table gives the variations of periodic current over a period

t	0	$T/6$	$T/3$	$T/2$	$2T/3$	$5T/6$	T
A	1.98	1.30	1.05	1.30	-0.88	-0.25	1.98

Show by numerical analysis, that there is a direct current part of 0.75 amp. In the variable current and obtain the amplitude of the first harmonic.

Solution: Hence length of the interval is T i.e., $l = \frac{T}{2}$

$$\text{Let } y = \frac{a_0}{2} + \left(a_1 \cos \frac{2\pi t}{T} + b_1 \sin \frac{2\pi t}{T} \right) + \dots,$$

where $\frac{a_0}{2}$ represents the direct current part and $\sqrt{a_1^2 + b_1^2}$ gives the amplitude of the first harmonic.

The values of t , A , $\sin \frac{2\pi t}{T}$, $\cos \frac{2\pi t}{T}$ are tabulated below:

t	A	$\cos \frac{2\pi t}{T}$	$\sin \frac{2\pi t}{T}$	$A \cos \frac{2\pi t}{T}$	$A \sin \frac{2\pi t}{T}$
0	1.98	1	0	1.98	0
$T/6$	1.30	0.5	$\frac{\sqrt{3}}{2}$	0.65	1.1258
$T/3$	1.05	-0.5	$\frac{\sqrt{3}}{2}$	-0.525	0.9093
$T/2$	1.30	-1	0	-1.30	0
$2T/3$	-0.88	-0.5	$-\frac{\sqrt{3}}{2}$	0.44	0.76208
$5T/6$	-0.25	0.5	$-\frac{\sqrt{3}}{2}$	-0.125	0.03125
Totals	4.5			1.12	2.82843

$$\text{We have } a_0 = \frac{2}{n} \sum A = \frac{2}{6} (4.5) = 1.50$$

$$a_1 = \frac{2}{n} \sum A \cos \frac{2\pi t}{T} = \frac{2}{6} (1.12) = 0.373, \quad b_1 = \frac{2}{n} \sum A \sin \frac{2\pi t}{T} = \frac{2}{6} (2.82843) = 1.005$$

$$\therefore A = 0.75 + 0.373 \cos \frac{2\pi t}{T} + 1.005 \sin \frac{2\pi t}{T} + \dots$$

$$\text{Direct current part} = \frac{a_0}{2} = 0.75 \text{ amp}$$

$$\text{The amplitude of the first harmonic} = \sqrt{a_1^2 + b_1^2} = \sqrt{(0.373)^2 + (1.005)^2} = 1.072$$

5. The turning moment T units of crank shaft of a steam engine is given for a series of values of the crank-angle θ in degrees:

θ	0	30	60	90	120	150	180
T	0	5224	8097	7850	5499	2626	0

Find the first four terms in a series of sines to represent T . Also calculate T when $\theta = 75^\circ$

Solution: Let the half range sine series to represent T be
 $T = b_1 \sin \theta + b_2 \sin 2\theta + b_3 \sin 3\theta + b_4 \sin 4\theta$

The values of θ , T, $\sin \theta$, $\sin 2\theta$, $\sin 3\theta$, $\sin 4\theta$ are tabulated below:

The values of t, A, $\sin \frac{2\pi t}{T}$, $\cos \frac{2\pi t}{T}$ are tabulated below:

θ	T	$\sin \theta$	$\sin 2\theta$	$\sin 3\theta$	$\sin 4\theta$
0	0	0	0	0	0
30	5224	0.5	0.866	1	0.866
60	8097	0.866	0.866	0	-0.866
90	7850	1	0	-1	0
120	5499	0.866	-0.866	0	0.866
150	2626	0.5	-0.866	1	-0.866

Using the values in the above table, we have

$$b_1 = \frac{2}{n} \sum T \sin \theta = \frac{2}{6} (23549) = 7850$$

$$b_2 = \frac{2}{n} \sum T \sin 2\theta = \frac{2}{6} (4499.736) = 1500$$

$$b_3 = \frac{2}{n} \sum T \sin 3\theta = \frac{2}{6} (0) = 0$$

$$b_4 = \frac{2}{n} \sum T \sin 4\theta = \frac{2}{6} (0) = 0$$

$$\therefore T = 7850 \sin \theta + 1500 \sin 2\theta$$

When $\theta = 75^\circ$, $T = 8322$

Exercise:

- Obtain the constant term and the first three coefficients in the Fourier series for y
Using the following table:

x	0	1	2	3	4	5
y	4	8	15	7	6	2

$$\text{Ans: } \frac{a_0}{2} = 7, a_1 = 4.565, a_2 = -2.833, a_3 = -1.667$$

- The following data gives the variation of periodic current over a period

t	0	$T/6$	$T/3$	$T/2$	$2T/3$	$5T/6$	T
A	9.0	18.2	24.4	27.8	27.5	22.0	9.0

Find numerically the direct current part of the variable current and obtain the amplitudes to the second harmonic.

$$\text{Ans: direct current part} = \frac{a_0}{2} = 21.4835$$

$$\text{Amplitude of the first harmonic} = 8.455$$

$$\text{Amplitude of the second harmonic} = 3.09$$



3. Obtain the the constant term and the first three coefficients of $\sin\theta$ and $\sin 2\theta$ in the Fourier expansion of y , using the following table:

θ°	0	60		120	180	240	300	360
y	0	9.2		14.4	17.8	17.3	11.7	0

Ans: $b_1 = -1.5588, b_2 = 0.1155$

4. Analyze the following data up to first harmonic

x	0	1	2	3	4	5	6	7	8	9	10	11
y	7	9	11	13	14	8	-7	-9	-11	-13	-14	-8

Ans: $0 + [2.122 \cos \frac{\pi x}{6} + 14.38 \sin \frac{\pi x}{6}]$

Video Links:

Links here

1. <https://nptel.ac.in/courses/111103021/15>
2. <https://www.youtube.com/watch?v=U8riFeiiu3s>
3. <https://www.youtube.com/watch?v=qhrfTPwBhBA>
4. https://www.youtube.com/watch?v=bzyBy_TWaTw
5. <https://www.youtube.com/watch?v=lDB7iEHwklg>